DS, CE, AI 01/02/23 - 10:00 am

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT ENGINEERING / SCHOOL OF TECHNOLOGY MANAGEMENT ENGINEERING

Program: B.Tech / MBA Tech (All Stream) / B.Tech Integrated (Civil) Year: II / III / IV Semester : III / V / VII

AcademicYear: 2022-2023

Subject: Management Accounting for Engineers

Marks: 100

Time: 10.00 am - 1.00 pm

Date : 01 February 2023

Durations: 3 (Hrs)

No. of Pages: 4

Re-Examination (2021-22 / 2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

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Q.1		**Answer briefly and show detailed calculations.**	20
CO- 1; SO- ; BL- Remember	a	Explain any FIVE accounting concepts with examples.	5
CO-4 ; SO- ; BL- Evaluate	b	A product is sold at Rs. 80 per unit. Its variable cost per unit is Rs 60, fixed cost is Rs. 6,00,000. Compute the followings: (1) P/V Ratio (2) Break-even point (3) Margin of safety at a sale of 50,000 units. (4) At what sale the producer will earn profit at 15% on sales.	5
CO- 4 ; SO- ; BL- Evaluate	c	The standard material cost for 200 units of output is : Material | Kg | Rate per Kg. A | 50 | 12 B | 100 | 9 C | 100 | 10 The actual cost for 8,000 units is as follows : Material | Kg | Rate per Kg. A | 2100 | 28,350 B | 3750 | 30,750 C | 4150 | 46,480 Calculate material cost variance, material price variance and material usage variance.	5

CO-4 ; SO- ; BL- Apply | d | 5

From the following particulars, calculate Reorder level, Maximum level, Minimum level.

Normal consumption	300 units per day
Maximum consumption	420 units per day
Minimum consumption	240 units per day
Re-order period	10 – 15 days
Re-order quantity	3,600 units
Normal re-order period	12 days

Q.2 CO-1; SO-; BL- Apply | a | 15

Prepare a cost sheet from the following particulars in the books of Bharati Ltd.

Raw material purchased	Rs.250000
Carriage Inward	Rs.20000
Direct wages	Rs.380000
Direct chargeable expenses	Rs.25000
Factory cost	25% of Prime cost
General and administrative expenses	5% of Factory cost
Selling and distribution expenses	10% of production cost
Profit	20% on sales

	Opening stock	**Closing stock**
Raw material	Rs.65000	Rs.80000
Work in progress	Rs.20000	Rs.28000
Finished goods	Rs.90000	Rs.110000

Q.2 CO-2; SO-; BL- Understand | b | 5

Explain Cost classification on the basis of Variability with examples.

Q.3 CO-3; SO-; BL-Apply | a | 15

A firm expects to have Rs. 25,000 in hand on 1st May 2022 and requires you to prepare a Cash Budget to estimate the cash position for the period of three months May, June and July 2022. The following information is supplied to you:

Months	**Sales (₹)**	**Purchases (₹)**	**Wages (₹)**	**Factory Expenses (₹)**	**Office Expenses (₹)**	**Selling Expenses (₹)**
March	50,000	30,000	6,000	5,000	4,000	3,000
April	56,000	32,000	6,500	5,500	4,000	3,000
May	60,000	35,000	7,000	6,000	4,000	3,500
June	80,000	40,000	9,000	7,500	4,000	4,500
July	90,000	40,000	9,500	8,000	4,000	4,500

Other Information:

a) 20% of sales is for cash and credit allowed to customers is one month,
b) Suppliers supply goods at 2 months credit,
c) All expenses are paid in the month following the one in which they are incurred,
d) Income-tax ₹ 25,000 is payable in July,
e) Plant has been ordered and is expected to be received in June and it will cost ₹ 80,000, and
f) The company pays dividends to shareholders and bonus to worker of ₹ 10,000 and ₹ 15,000 respectively in the month of May.

<table>
<tr><td>Q.3
CO-3 SO-;
BL-
Understand</td><td>b</td><td>Differentiate between Fixed Budget and Flexible Budget.</td><td>5</td></tr>
<tr><td>Q.4
CO- 5; SO-;
BL-Apply</td><td>a</td><td>The Cost sheet of Alpha ltd is as follows :
<table><tr><td></td><td>Product A (Rs)</td><td>Product B (Rs.)</td><td>Product C(Rs.)</td></tr><tr><td>Sales</td><td>112000</td><td>56000</td><td>84000</td></tr><tr><td>Direct Material</td><td>14000</td><td>7000</td><td>14000</td></tr><tr><td>Direct Labour</td><td>5600</td><td>7000</td><td>22400</td></tr><tr><td>Variable overheads</td><td>14000</td><td>7000</td><td>28000</td></tr><tr><td>Fixed Cost</td><td>28000</td><td>14000</td><td>28000</td></tr><tr><td>Total Cost</td><td>61600</td><td>35000</td><td>92400</td></tr></table>The management is considering to discontinuing Product C. There is no possibility of reducing fixed costs. Advise whether Product C should be discontinued or not.</td><td>10</td></tr>
<tr><td>Q.4
CO- 4; SO-;
BL- Evaluate</td><td>b</td><td>Raj Ltd. provides the following data :
<table><tr><td>Particulars</td><td>2021 (First Half)</td><td>2021 (Second Half)</td></tr><tr><td>Sales (Rs.)</td><td>405000</td><td>513000</td></tr><tr><td>Profit (Rs.)</td><td>10800</td><td>32400</td></tr></table>From the above, calculate PV ratio, Break-even point in Rupees, and Fixed cost. Also find out profit or loss where sales are Rs.324000 and Sales required to earn profit of Rs.54000.</td><td>10</td></tr>
<tr><td>Q.5
CO-2; SO-;
BL-Apply</td><td>a</td><td>The standard labour budget of a company for a month is as follows :
20 skilled men @ 50 paise per hour for 40 hours = Rs 400
40 unskilled men @ 30 paise per hour for 40 hours = Rs 480
Total = Rs 880
The actual labour force was used as follows :
30 Skilled men @ 50 paise per hour for 40 hours = Rs 600
30 Unskilled men @ 35 paise per hour for 40 hours = Rs 420
Total = Rs 1,020
Calculate the labour cost, rate, and efficiency variances.</td><td>12</td></tr>
<tr><td>Q.5
CO-1; SO-;
BL-
Remember</td><td>b</td><td>Distinguish between Activity Based Costing and Traditional Costing.</td><td>8</td></tr>
</table>

Q.6 CO-4, SO-; BL- Analyse	a	A firm sells 25000 units at a selling price of Rs. 5 per unit. Its fixed cost is Rs. 40,000 and variable expenses Rs. 50,000. Find out the break-even point for the firm. Also, find out BEP when, 1) The selling price is increased by 30% 2) The fixed cost is increased by 15% 3) The fixed cost is decreased by 25% 4) The selling price is deceased by 20%	15
Q.6 CO-1; SO-; BL- Remember	b	Explain advantages of Cost accounting.	5
Q.7 CO-3; SO-; BL- Analyse	a	The expenses budgeted for production of 10,000 units in a factory are furnished below : **Cost Per Unit (₹)** Direct materials — 70 Direct labour — 25 Variable factory overheads — 20 Fixed factory overheads (₹ 1,00,000) — 10 Direct variable expenses — 5 Selling expenses (10% fixed) — 13 Distribution expenses (20% fixed) — 7 Administrative expenses (₹ 50,000) — 5 **Total Cost Per Unit** — **155** You are required to prepare a flexible budget for the production of 8,000 units.	15
Q.7 CO-2; SO-; BL- Evaluate	b	Exide Ltd. requires 2,000 kgs of raw material for the production of "Z" per month. The cost of placing an order is Rs. 240. The purchase price of the raw material is Rs. 10 per kg. The carrying cost of inventory is 20% per annum. Calculate Economic Order Quantity. Also, determine number of orders to be placed during the year.	5

THE END